

SEFS LAB HPC COMMAND CHEAT SHEET

General command guide for NC State HPC (Hazel)

Cluster access: `ssh <unityid>@login.hpc.ncsu.edu` | OS: RHEL 9.2

Author: Michelle Pretorius | Updated: Aug 2026 | Scheduler: Slurm

1. FILESYSTEMS & STORAGE DIRECTORIES

Location	Path & Quota	Purpose / Policy
Home	<code>~/</code> or <code>/home/<unityid></code> 1 GB QUOTA	Scripts, small apps. Backed up daily.
Scratch	<code>/share/doserlab/<unityid></code> 20 TB QUOTA	Job execution, raw results. Not backed up! Purged after 30 days inactive.
Apps	<code>/usr/local/usrapps/doserlab/jwdoser</code>	SEFS Lab shared software and R packages repository.
Research	<code>/rs1</code> or <code>/rsstu</code> 2TB – 30TB	Long-term project data storage across nodes.

2. CONNECTION & INTERACTIVE SFTP

Command	Description
<code>ssh <unityid>@login.hpc.ncsu.edu</code>	Log in to Hazel via SSH (requires Duo 2FA).
<code>sftp <unityid>@login.hpc.ncsu.edu</code>	Open interactive secure file transfer session.
<code>exit</code> / <code>Ctrl+D</code>	Terminate SSH or SFTP connection.
<code>hostname</code>	Identify current node (<code>loginXX</code> vs compute node).
<code>whoami</code>	Print active UnityID.
<code>clear</code>	Clear terminal screen.

Interactive SFTP sub-commands (inside `sftp` prompt):

<code>put <local> <remote></code>	Upload file to HPC (add <code>-r</code> for folders).
<code>get <remote> <local></code>	Download file from HPC (add <code>-r</code> for folders).
<code>lcd <path></code>	Change working directory on local machine .
<code>lpwd</code>	Print local working directory.

3. DIRECTORY & FILE OPERATIONS

Command	Description / Key Options
<code>pwd</code>	Print current full working directory path.
<code>cd <path></code>	Change directory (<code>cd ~</code> home, <code>cd ..</code> parent).
<code>ls -ltr</code>	List files in long format sorted by modification time.
<code>mkdir <folder></code>	Create a new directory (<code>rmdir</code> removes empty dir).
<code>nano <file></code>	Command-line text editor (<code>Ctrl+X</code> to exit/save).
<code>cat <file></code>	Print entire file contents to terminal screen.
<code>head -n 20 <f></code>	Print first 20 lines of a file.
<code>tail -f <f></code>	Print last 10 lines of a file (<code>-f</code> monitors live).
<code>less <file></code>	Page through file line-by-line (press <code>q</code> to exit).
<code>grep "str" <f></code>	Search pattern inside file (e.g. <code>ls grep txt</code>).
<code>cp -r <s> <d></code>	Copy file or directory (<code>-r</code> recursive).
<code>mv <s> <d></code>	Move or rename a file or directory.
<code>rm <file></code>	Permanently delete file (Warning: cannot undo!).

4. QUOTA & MODULE MANAGEMENT

Command	Description
<code>quota -s</code>	Display home directory quota and space used.
<code>du -sh .</code>	Summarize total space used by current directory.
<code>module avail</code>	List all software modules available on Hazel.
<code>module load R</code>	Load specific module into active session (e.g., R).
<code>module list</code>	Show software modules currently loaded.
<code>module purge</code>	Unload all modules (best practice at start of scripts).

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5. SLURM BATCH SCRIPT DIRECTIVES (#SBATCH)

Directive Flag	Description / Example
<code>-job-name=<name></code>	Name displayed in queue (e.g., <code>myjob</code>).
<code>-output=stdout.%x.%j</code>	Standard output log file (<code>%x</code> = job name, <code>%j</code> = job ID).
<code>-error=stderr.%x.%j</code>	Standard error log file.
<code>-ntasks=1</code>	Number of tasks/processes (MPI ranks).
<code>-cpus-per-task=4</code>	Number of CPU cores per task (threads).
<code>-nodes=1</code>	Number of compute nodes requested.
<code>-time=02:00:00</code>	Wall clock limit (<code>HH:MM:SS</code> or <code>D-HH:MM:SS</code>).
<code>-mem=16G</code>	Memory per node (e.g., <code>16G</code> or <code>16000M</code>).
<code>-partition=compute</code>	Partition selection: <code>compute</code> (CPU) or <code>gpu</code> .
<code>-qos=normal</code>	QOS tier: <code>normal</code> (max 4 days) or <code>long</code> (max 10 days).
<code>-array=1-10</code>	Submit job array with task indices (e.g., 1, 2, ... 10).

6. TEMPLATE SLURM BATCH SCRIPT

```
#!/bin/bash
#SBATCH --job-name=myjob
#SBATCH --output=stdout.%x.%j
#SBATCH --error=stderr.%x.%j
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4
#SBATCH --time=02:00:00
#SBATCH --mem=16G
#SBATCH --partition=compute
#SBATCH --qos=normal

# Explicitly switch to submission directory
cd $SLURM_SUBMIT_DIR

# Clean environment & load modules
module purge
module load R

# Execute R script
Rscript ~/path/to/script/my_model.R
```

7. SLURM JOB MANAGEMENT COMMANDS

Command	Description
<code>sbatch submit.sh</code>	Submit batch script to Slurm execution queue.
<code>salloc</code>	Start an interactive session on a compute node.
<code>srun <exec></code>	Launch parallel job step (MPI) or command.
<code>squeue</code>	View status (<code>ST</code>) of your queued/running jobs.
<code>sjs <JOBID></code>	View live CPU %, RAM, and disk I/O for running job.
<code>scancel <JOBID></code>	Cancel running job (<code>scancel -u \$USER</code> cancels all).
<code>sacct -j <JOBID></code>	View accounting details/exit state for completed job.
<code>seff <JOBID></code>	Display CPU and memory efficiency utilization report.
<code>sa</code>	Show user account allocations and allowed QOS.
<code>sqos</code>	View QOS policies, priorities, and max wall time.
<code>si</code>	Show compute node availability across partitions.

8. SLURM STATES & ENVIRONMENT VARIABLES

Code	Status	Description
<code>PD</code>	Pending	Waiting for requested cluster resources or QOS slot.
<code>R</code>	Running	Job is actively executing on allocated compute node(s).
<code>CD</code>	Completed	Job finished successfully with exit code 0.
<code>F</code>	Failed	Non-zero exit code (check <code>stderr</code> for exact errors)
<code>TO</code>	Timeout	Exceeded requested <code>-time</code> limit.
<code>OOM</code>	Out of memory	Job process was killed for exceeding <code>-mem</code> limit.

Slurm environment variables (used inside scripts):

<code>\$SLURM_SUBMIT_DIR</code>	Directory path where <code>sbatch</code> was executed.
<code>\$SLURM_ARRAY_TASK_ID</code>	Current task index for Job Array (e.g., 1, 2, 3...).
<code>\$SLURM_JOB_ID</code>	Unique numeric ID assigned to the job (same as <code>%j</code>).